Icepi Zero Testing Guide

Setup

Download the following files:

https://github.com/cheyao/icepi-zero/raw/refs/heads/main/documentation/fire-v.bit

https://raw.githack.com/cheyao/icepi-zero/main/documentation/Icepi-Zero-FTDI-Chip.xml, press Ctrl+S to download. (https://github.com/cheyao/icepi-zero/blob/main/documentation/Icepi-Zero-FTDI-Chip.xml)

Install the following applications:

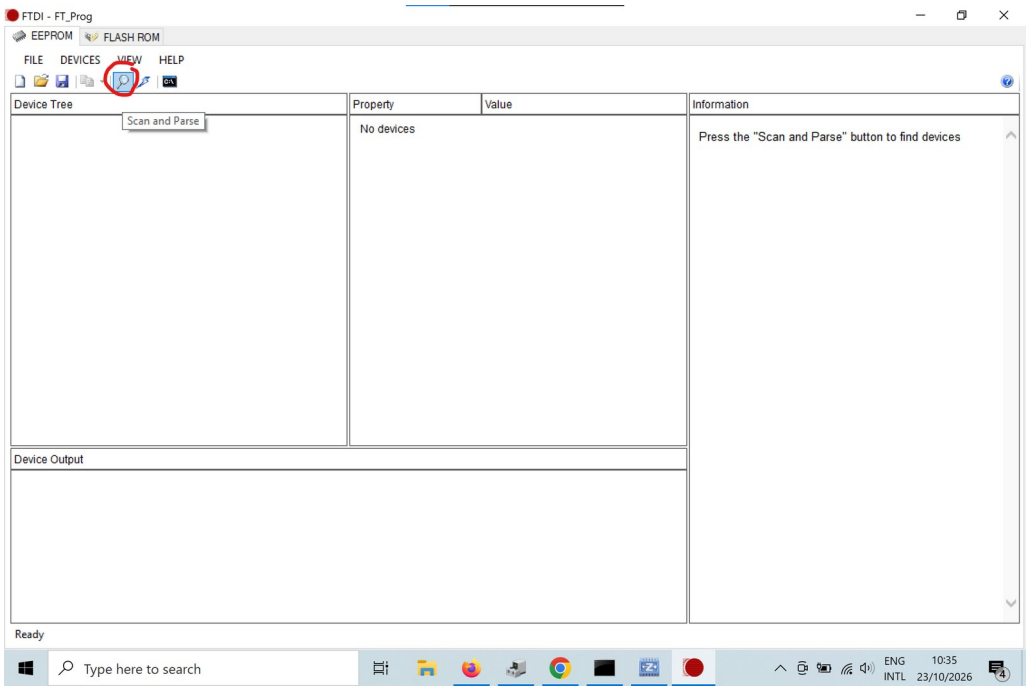
https://github.com/pbatard/libwdi/releases/download/v1.5.1/zadig-2.9.exe (Zadig USB utility from https://zadig.akeo.ie/)

https://ftdichip.com/wp-content/uploads/2026/07/FT_Prog_v3.12.82.707-Installer.zip (FT_Prog from https://ftdichip.com/utilities/)

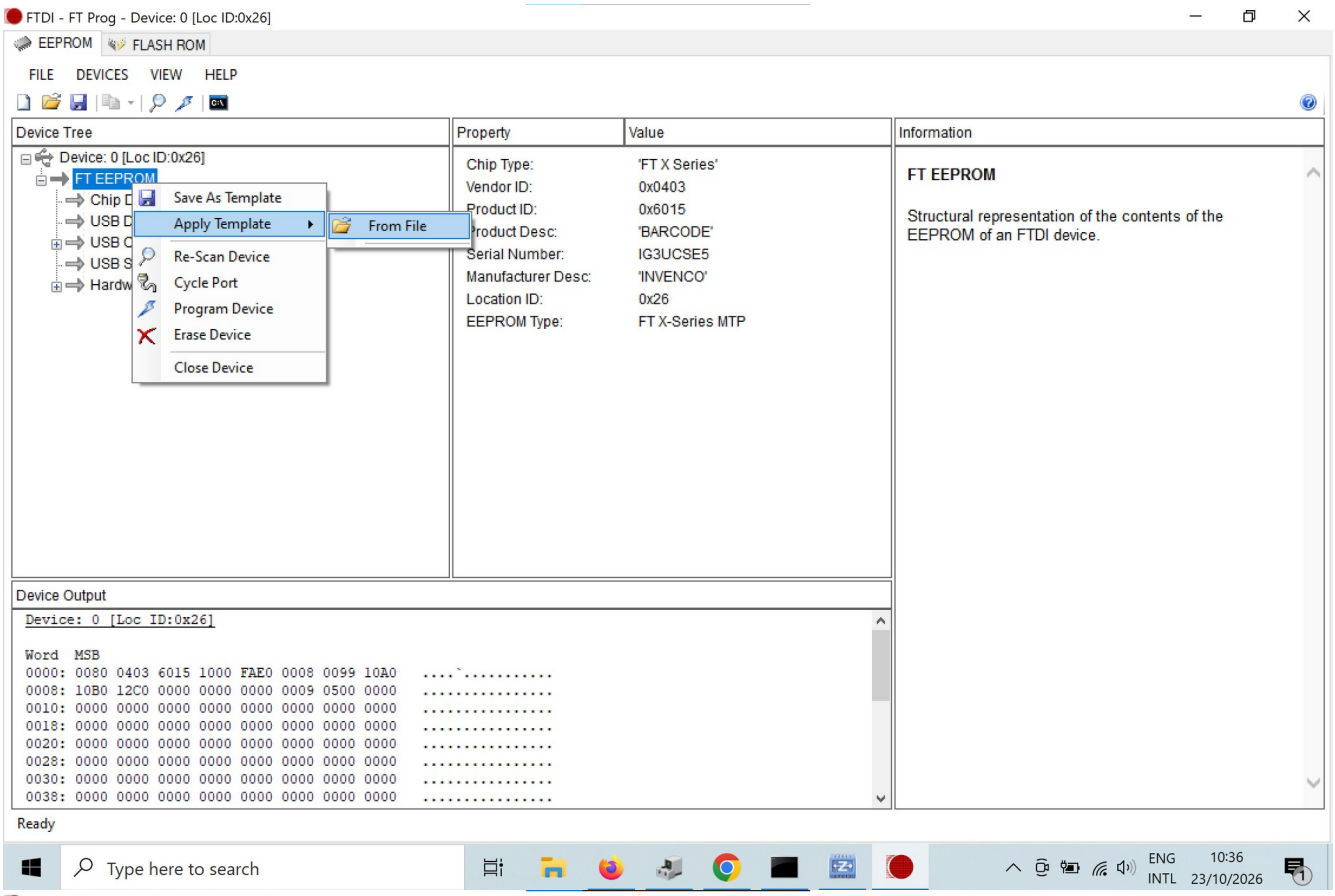
Testing

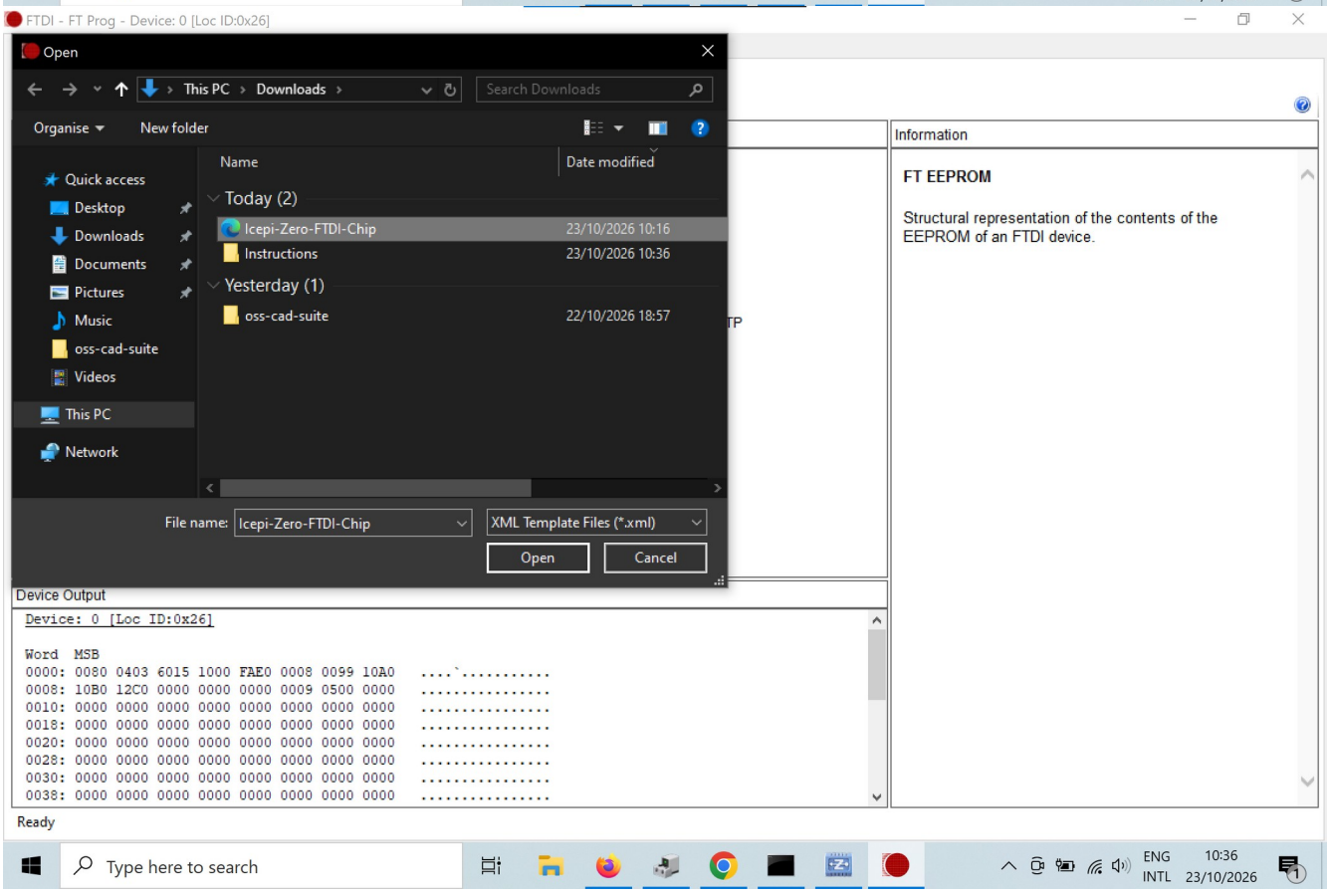
1. Plug in a USB-C data cable to the left-most USB port on the PCB. (The one next to the HDMI port labeled Flash). Connect the other end to a Windows PC/laptop. Also use a micro-HDMI to HDMI cable to connect the HDMI port to a monitor.

2. Open FT_Prog app and click “Scan and Parse”

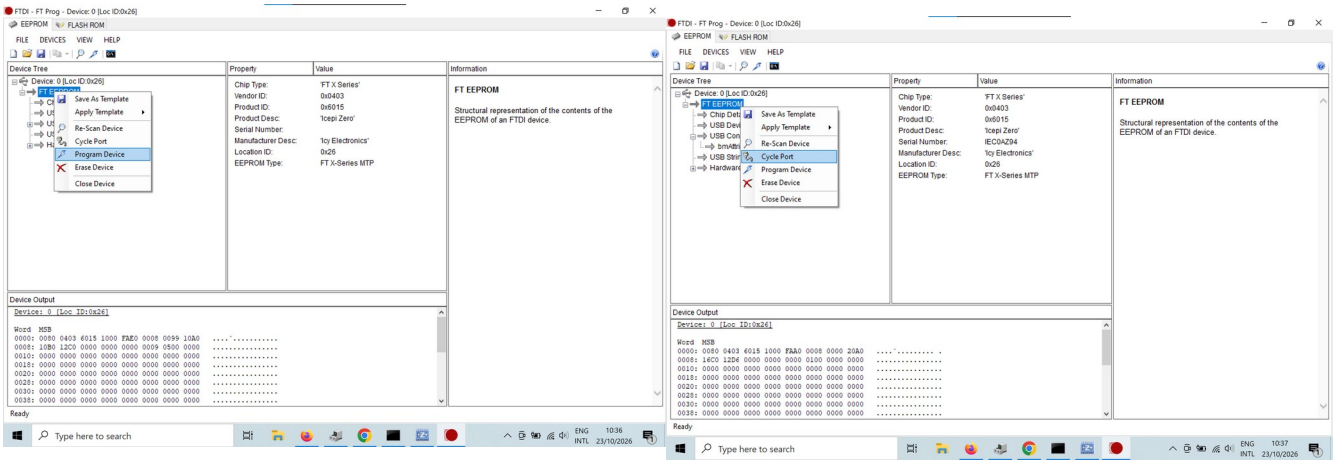


3. Right click the newly appeared device and click Apply Template > From File and select the provided “Icepi-Zero-FTDI-Chip.xml”

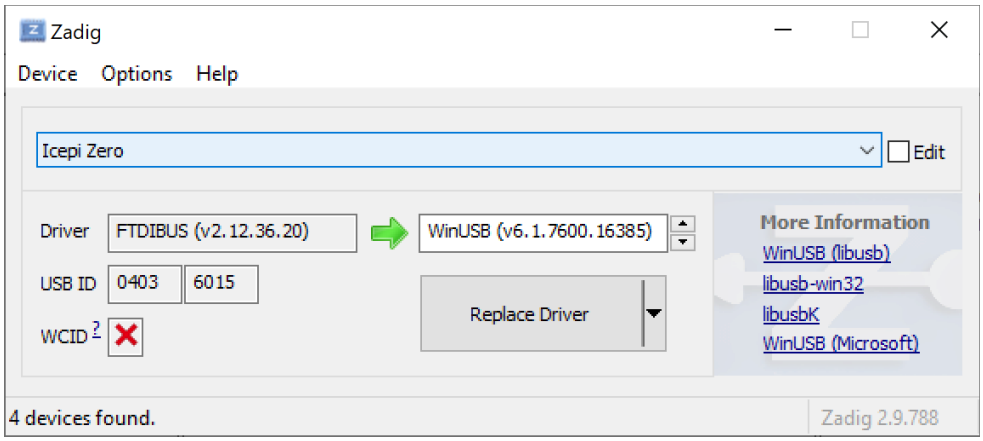

FTDI - FT Prog - Device: 0 [Loc ID:0x26]
EEPROM
FLASH ROM
FILE
DEVICES
VIEW
HELP
Device Tree
Device: 0 [Loc ID:0x26]
FT EEPROM
Chip D
USB D
USB C
USB S
Hardw
Save As Template
Apply Template
From File
Re-Scan Device
Cycle Port
Program Device
Erase Device
Close Device
Property
Value
Chip Type:
'FT X Series'
Vendor ID:
0x0403
Product ID:
0x6015
Product Desc:
'BARCODE'
Serial Number:
IG3UCSE5
Manufacturer Desc:
'INVENCO'
Location ID:
0x26
EEPROM Type:
FT X-Series MTP
Information
FT EEPROM
Structural representation of the contents of the EEPROM of an FTDI device.
Device Output
Device: 0 [Loc ID:0x26]
Word MSB
0000: 0080 0403 6015 1000 FAE0 0008 0099 10A0 ....`...........
0008: 10B0 12C0 0000 0000 0000 0009 0500 0000 ....
0010: 0000 0000 0000 0000 0000 0000 0000 0000 ....
0018: 0000 0000 0000 0000 0000 0000 0000 0000 ....
0020: 0000 0000 0000 0000 0000 0000 0000 0000 ....
0028: 0000 0000 0000 0000 0000 0000 0000 0000 ....
0030: 0000 0000 0000 0000 0000 0000 0000 0000 ....
0038: 0000 0000 0000 0000 0000 0000 0000 0000 ....
Ready
Type here to search
ENG
INTL
10:36
23/10/2026


FTDI - FT Prog - Device: 0 [Loc ID:0x26]
Open
This PC
Downloads
Search Downloads
Organise
New folder
Quick access
Desktop
Downloads
Documents
Pictures
Music
oss-cad-suite
Videos
This PC
Network
Name
Date modified
Today (2)
Icepi-Zero-FTDI-Chip
23/10/2026 10:16
Instructions
23/10/2026 10:36
Yesterday (1)
oss-cad-suite
22/10/2026 18:57
File name:
Icepi-Zero-FTDI-Chip
XML Template Files (*.xml)
Open
Cancel
Information
FT EEPROM
Structural representation of the contents of the EEPROM of an FTDI device.
Device Output
Device: 0 [Loc ID:0x26]
Word MSB
0000: 0080 0403 6015 1000 FAE0 0008 0099 10A0 ....`...........
0008: 10B0 12C0 0000 0000 0000 0009 0500 0000 ....
0010: 0000 0000 0000 0000 0000 0000 0000 0000 ....
0018: 0000 0000 0000 0000 0000 0000 0000 0000 ....
0020: 0000 0000 0000 0000 0000 0000 0000 0000 ....
0028: 0000 0000 0000 0000 0000 0000 0000 0000 ....
0030: 0000 0000 0000 0000 0000 0000 0000 0000 ....
0038: 0000 0000 0000 0000 0000 0000 0000 0000 ....
Ready
Type here to search
ENG
INTL
10:36
23/10/2026

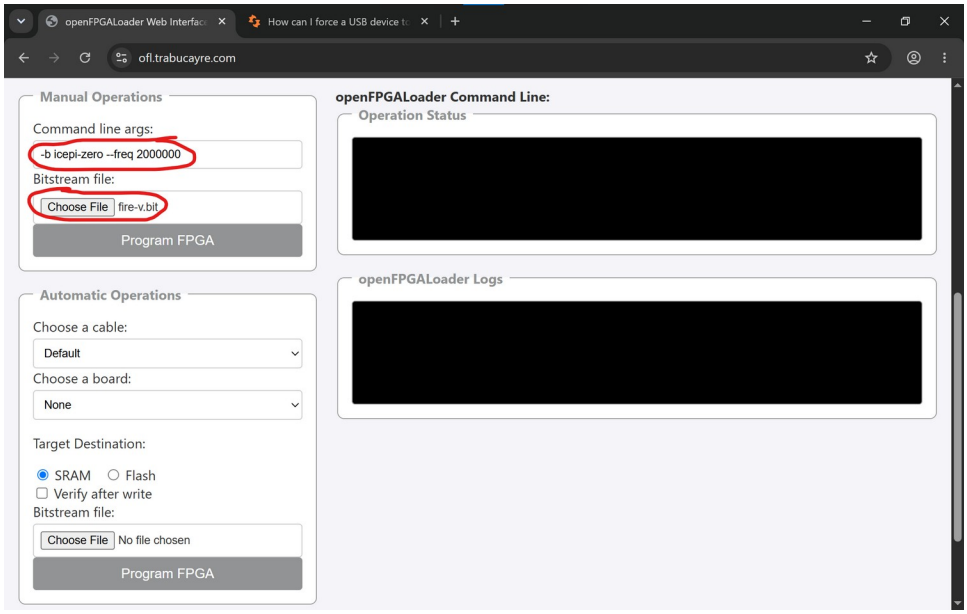
4. Right click the FT EEPROM device and click program device, then cycle port



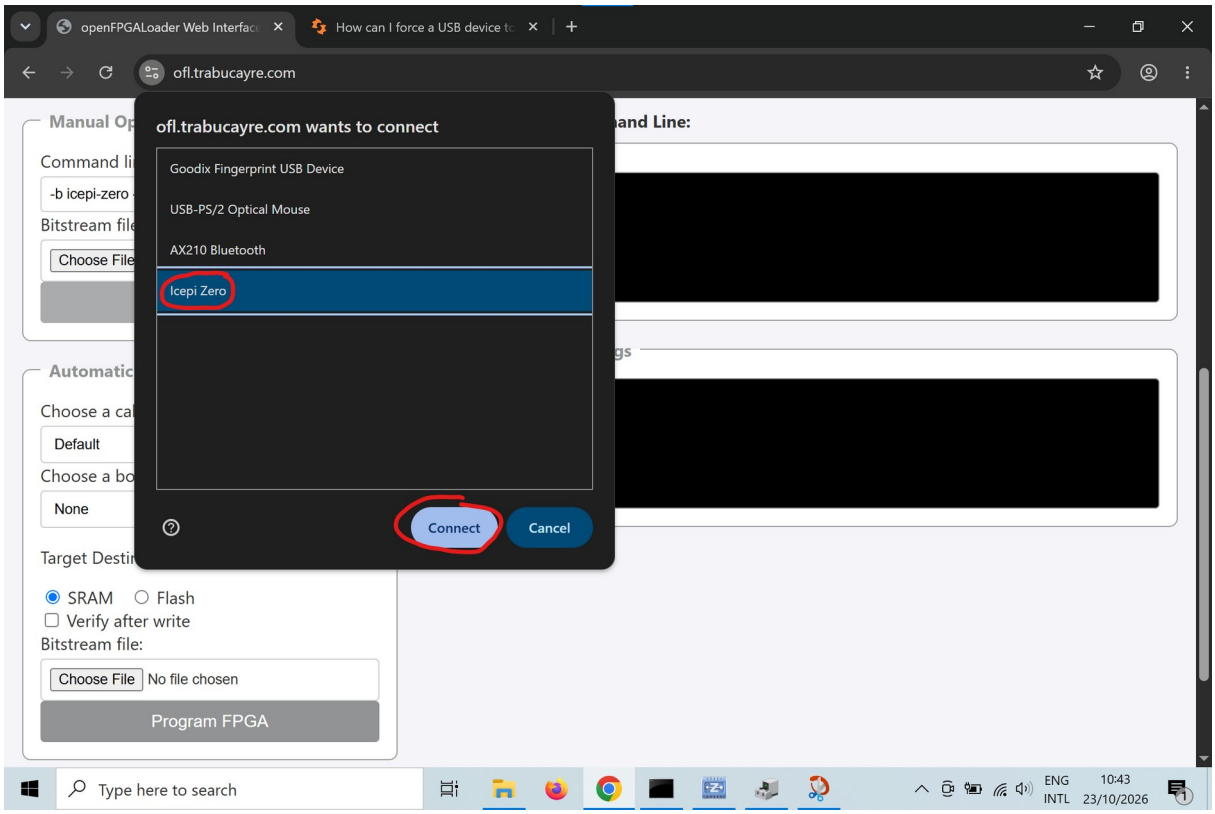
5. Open Zadig, select Icepi Zero (You may need to go into Options and check show all USB devices), then select WinUSB on the right side of the green arrow. Click Replace Driver. This might take 20-30 seconds



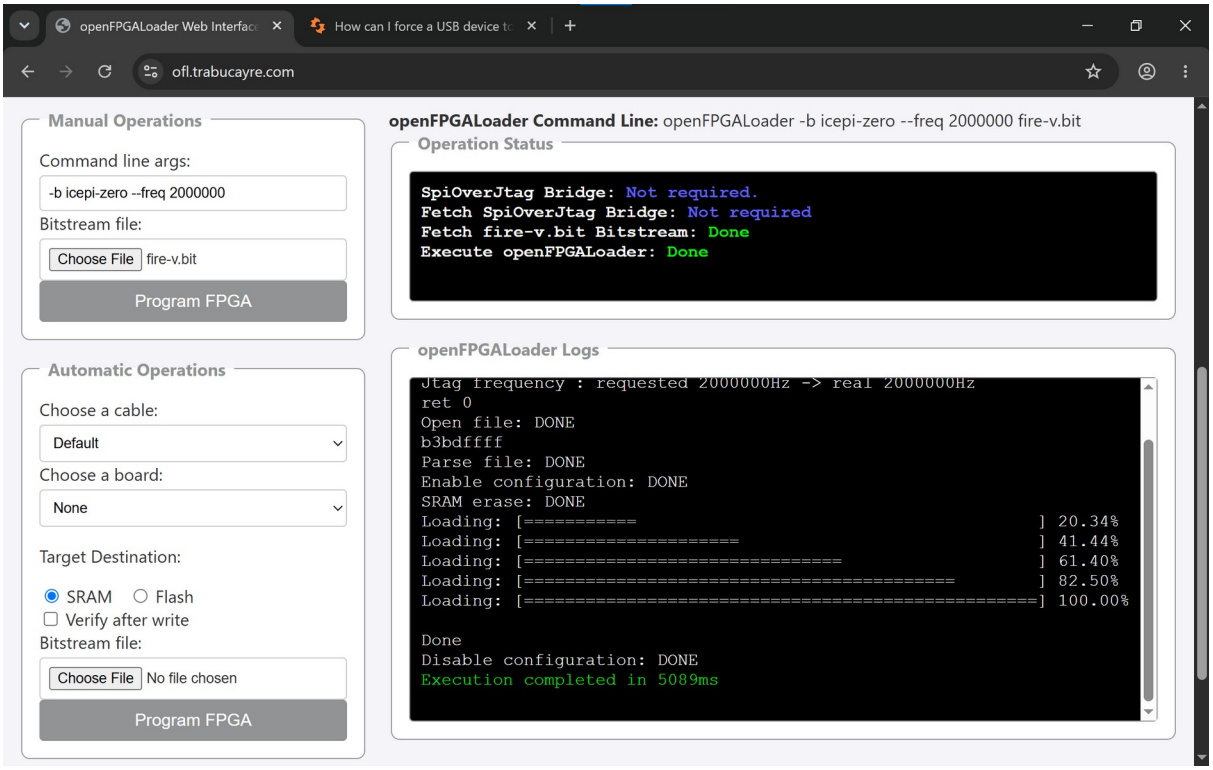
6. Open https://ofl.trabucayre.com/ in Google Chrome (Firefox does **NOT** work). On the middle left, in the “Manual Operations” section, select the following arguments (Click “Choose File” and select the fire-v.bit file downloaded during setup)



7. Click on “Program FPGA” in the Manual Operations section and select “Icepi Zero” in the popup menu and click connect:



8. Wait for a few seconds, “Done” should be written in “openFPGALoader Logs”:



7. If the PCB works, you should see all the LEDs blinking and the following HDMI output on the monitor:

